

qicert: Certified Quantum-Inspired Compression of Vision–Language–Action Models with Exact Bounds, Provable Pruning, and Formal Safety Tails

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Code: <https://github.com/ZakLr/qicert> Reproduces every table via `qicert.bench.all`

Abstract—Vision–Language–Action Models (VLAMs) at 7B–10B parameters are too large for embedded vehicle and robot controllers and too expensive to certify. We present *qicert*, a backbone-agnostic pipeline that compresses VLAMs into quantized tensor-train (QTT) networks before the heaviest training, compiles cross-modal interaction blocks into exactly-evaluable commuting-Pauli families, and certifies what remains with a three-layer certificate stack: exact algebraic bounds (Lipschitz products of QTT cores; per-family pruning errors), local formal proofs (SOS boxes; Lyapunov-margin survival curves), and statistical tail bounds on signal-temporal-logic robustness estimated by iterative quantum amplitude estimation raced against the strongest honest classical estimator (RESTART+GEV) and closed by scenario-optimization and conformal calibration. A syndrome-shadow monitor guards the action stream with theorem-budgeted false-alarm rates. We report the honest measured status on a fine-tuned MiniVLA-1B (full frozen 108-episode split, validated harness, FT-anchor exactness +0.0000): calibrated INT8 is accuracy leader at $2\times$ compression (0.4466 vs 0.4468 dense), while our certified TT+residual track measurably trails on accuracy at $2.86\times$ (0.1529) — and is the only compressed model in the comparison that ships sound runtime certificates, enforced by a load/step guard that refuses every action or artifact it cannot certify. The INT8 baseline cannot compute these certificates, which are exact functions of the compressed cores.

Index Terms—tensor networks, quantized tensor train, Lipschitz certificates, Pauli grouping, Clifford diagonalization, quantum amplitude estimation, signal temporal logic, classical shadows, VLAM, autonomous driving, robotics, formal safety

I. INTRODUCTION

VLAMs fuse camera, LiDAR, radar, and language inputs into planning and control outputs. Reference models such as Alpaymoy-R1 [1] and π_0 [2] demonstrate the category’s potential, but their scale makes them prohibitively expensive to train and nearly impossible to deploy on embedded hardware within the 100 ms real-time budget while satisfying functional-safety certification (ISO 26262; IEC 62061 / ISO 10218).

The challenge asks for a reproducible quantum-enhanced or quantum-inspired component with a clear advantage over a classical baseline across four bottlenecks: model footprint, attention complexity, RL alignment cost, and control-safety guarantees. Quantum-inspired tensor networks have demonstrated strong LLM compression [3], and provably efficient quantum algorithms promise gains in the high-dimensional linear algebra central to VLAM training [4]. Our answer is a

system: one pipeline in which five quantum-inspired primitives — tensor networks, Pauli observables, amplitude encoding, syndrome measurement, and shadow tomography — are the type system of every stage (train $\rightarrow$ compress $\rightarrow$ compile $\rightarrow$ certify $\rightarrow$ monitor). At the same time, the *classical-energy* tier (a MiniVLA-1B robotics baseline on the LIBERO spatial slice, with INT8 weight-only quantization + claws) is the cross-cutting fidelity and energy context for Q3: it tells us, honestly, whether the classical-quantized relationship is actually the interesting physics or whether quantum matters. The two tiers share one repository and one ledger but have separate hardware, separate stacks, and separate kill criteria; they are not the same experiment presented twice. The contributions are:

- 1) **Exact-certificate compression**: QTT cores double as the objects of an exact global Lipschitz bound — exact per-layer factors, residual gap measured (Sec. III) — giving a certified-safe-set column that no quantization-based baseline can produce at any compression ratio.
- 2) **Compilation-as-identity**: cross-modal interaction tensors are grouped into commuting Pauli families, simultaneously diagonalized by single Cliffords, and evaluated exactly; dropping a family costs an exact, printed error $\|\sum_{a \in F_j} c_a\|_1$ (Sec. IV).
- 3) **Quadratic query-complexity savings on the safety tail (realized oracle)**: the certified failure probability is estimated by iterative quantum amplitude estimation (IQAE) with Bayesian credible intervals, over a structured scenario domain with a closed-form certificate-margin oracle (Sec. V), raced end-to-end against RESTART+GEV at matched total cost and closed by Campi–Garatti scenario-optimization and conformal prediction.
- 4) **A runnable, self-reproducing artifact**: *qicert* (“certified quantum-inspired compression”) reproduces every table from a clean environment and emits its own hardware pathway (qubits, depth, shots).

All quantum components are simulated per challenge constraints (Sec. 5.3); resource assumptions are stated per algorithm. Results below are reported as pre-registered targets with measured values pending (Sec. VIII).

TABLE I

THE SEVEN FUSION JOINTS. EACH COMPONENT CONSUMES ANOTHER’S OUTPUT, SO ABLATIONS ISOLATE CLEANLY AND THE MARGINAL COST OF EVERY EXTRA CLAIM IS NEAR ZERO.

Joint	What it buys
1	QTT cores = certificate inputs (zero extra compute)
2	Compiled families = monitor’s measurement basis (zero extra design)
3	Certificate margins = the STL predicates IQAE estimates (compute multiplicity)
4	Sec. 5.2 seeded scenarios = the amplitude-estimation oracle
5	Worst-case long-tail margin = bond-allocation objective (tail designs model)
6	Compiled qubit/depth/shots table = the hardware pathway
7	qicert carries everything

Stage 0: baseline fine-tune (QLoRA/LoRA)

- ↓ Stage 1: {TT, QTT} TT-cross compression + bond allocator
- ↓ Stage 2: ALS/DMRG-native fine-tuning (dense matrix never formed)
- ↓ Stage 3: compile commuting-Pauli families (exact, prunable)
 - ↓ Stage 4: certify — L1 exact Lipschitz / L2 SOS + Lyapunov / L3 IQAE tails
- ↓ Stage 5: package qicert; bench suite reproduces every table

Fig. 1. Training-time pipeline (Sec. II); the runtime view adds the syndrome-shadow monitor (Sec. V-D).

II. SYSTEM ARCHITECTURE

The pipeline is identical across both tracks; only the slice specification differs.

Training time (offline, Kaggle 2×T4): **Stage 0** baseline fine-tune (QLoRA/LoRA) of the backbone (AD: LLaVA-1.5-7B with Alpamayo-adjacent action head, justified per Sec. 5.4; robotics: MiniVLA-class head). **Stage 1** compress into {TT, QTT} via TT-cross (query-based, no global SVD); the safety-budgeted bond allocator sets per-layer ranks. **Stage 2** ALS/DMRG-native fine-tuning on the compressed manifold — the dense matrix is never formed. **Stage 3** compile cross-modal interaction blocks into commuting-Pauli families. **Stage 4** certify (three layers, Sec. III). **Stage 5** package into qicert; the bench suite reproduces every table.

Runtime (deployed, ≤100 ms profile): sensor + instruction encoders → compiled interaction kernels (diagonal, $O(N \cdot k)$ -style) → QTT-compressed action head with certified margins → syndrome-shadow monitor (PDU hygiene + shadow statistics) → safe output or certified fallback action.

Fusion joints (why this is a system, not a portfolio):

Unifying principle. A VLAM that will be certified and deployed on embedded hardware is, at every stage, a computation on *structured tensors* — weights, interaction tensors, belief and robustness distributions. Quantum information is the discipline that made structured tensors a first-class programming model: tensor networks, Pauli observables, amplitude encoding, syndrome measurement, shadow tomography. We build the entire train → compress → compile → certify → monitor pipeline from those five primitives. Quantum-inspiration is the type system of the whole submission, not a label on one component.

III. THE CERTIFICATE STACK

Three layers answer three questions and are never confused.

A. Layer 1: Exact algebraic bounds

For a TT/QTT layer with cores $G_1 \dots G_d$ the reconstructed weight matrix is $W = \text{contract}(G_1, \dots, G_d)$, and norm submultiplicativity gives

$$\|W\|_2 \leq \prod_{i=1}^d \|G_i\|_2, \quad (1)$$

each core norm computed by power iteration on the small core in $O(\sum_i r_i^2 n_i)$. The bound is deliberately conservative; the *exact* per-layer operator norm is available at the same cost class: power iteration on the TT contraction performs matvecs $x \mapsto Wx$ in $O(\sum_i r_{i-1} n_i^2 r_i)$ per step (standard TT contraction order; $O(dn^2 r^2)$ at uniform mode size) without ever forming W . Chaining through the network with 1-Lipschitz activations (ReLU et al.),

$$L(\tilde{F}) = \prod_{\text{layers } l} \|W^{(l)}\|_2, \quad (2)$$

where each factor is the exact per-layer norm (power iteration), not the core-product bound. The residual gap between $L(\tilde{F})$ and $\sup_x \|\nabla \tilde{F}(x)\|_2$ — activation interleaving and non-orthogonality across layers — is *measured*, not assumed: N3 registers the tightness ratio $\kappa = L(\tilde{F})/\hat{L}_\nabla$, where $\hat{L}_\nabla$ estimates $\sup_{x \in B} \|\nabla \tilde{F}(x)\|_2$ by power iteration on the full TT network at adversarial sample points in B , and prints it at every Pareto point. With the input box B from the slice spec this yields the *exact safe-set certificate*

$$\text{Safe}(B) = \{x \in B : \|\tilde{F}(x) - \tilde{F}(x_{\text{ref}})\| \leq L(\tilde{F}) \text{diam}(B) < \text{margin}\}. \quad (3)$$

Non-emptiness is a *measured* column (Table V), never an assumption: a conservative L shrinks the column but cannot falsify it, and the column still separates us from INT8, for which no certificate object exists at any ratio. Its usefulness is gated by R2; where the global bound is loose, Layer 2a supplies the tight local boxes.

Theorem 1 (Lipschitz product, layer 1). *For a TT layer with cores $G_1 \dots G_d$, Eq. (1) holds with no sampling and no input-distribution assumption, and the tighter exact factor $\|W\|_2$ is computable by power iteration on the TT contraction at cost $O(k_{\text{it}} \sum_i r_{i-1} n_i^2 r_i)$ with k_{it} iterations, the dense matrix never formed. Both properties are preserved for QTT (log-mode) reshapes because they depend only on the contraction, not on mode sizes.*

The load-bearing comparison: INT8 quantization error is data-dependent (clip ranges, per-channel scales) and unfactorizable — no exact global constant exists. SVD truncation gives no certificate object to compose. Our compressed cores *are* the certificate.

[Measured in N3/N5: certified-safe-set % vs compression ratio]

qicert-QTT: monotone decay, bond-spectrum predictor band
INT8 / SVD: not computable (flat 0% line, stated honestly)

Fig. 2. The asymmetry plot (pre-registered): the certificate column that quantization baselines structurally cannot produce.

B. Layer 2: Local formal proofs

(a) SOS-certified boxes. Composition polynomials of TT-structured networks admit moment-SOS hierarchies with tractable SDPs [8]. SOS is deliberately *local by construction*: each certificate covers one sampled long-tail box B_k and a low-degree (≤ 4) composition polynomial over a small slice (state dimension $\lesssim 16$), never a whole-network SDP — formal verification of a 7B network in one SDP is out of scope for any method, and the hierarchy’s state-lifting structure keeps each box small. R3 budgets each box at <2 h and demotes SOS to a report line if violated. For each sampled long-tail box B_k we certify

$$\forall x \in B_k : \text{margin}(x) \geq m_k \geq 0, \quad (4)$$

proven by an exported SOS certificate (machine-checkable Gram matrix). Layer 1 is global-but-loose; Layer 2a is local-but-tight; disagreement between them is a surfaced bug, not a hidden number.

(b) Lyapunov-margin degradation curve + predictor. The old binary “certificate survives” claim is upgraded to a continuous object. At each compression ratio r ,

$$\text{survival}(r) = \frac{\#\{\text{trajectories where } V\text{-decrease holds}\}}{\#\{\text{trajectories}\}}, \quad (5)$$

with a regression from (bond spectra, entropy signals, layer sensitivities) to $\text{survival}(r)$. The submission prints the *predicted* largest safe ratio r^* *before* measuring it, then reports $|r^* - r_{\text{measured}}|$.

Measured form (N5-v2, this revision). The degradation curve is measured as token-level *action agreement* between each compressed checkpoint and the fine-tuned reference over a fixed prefix of the frozen eval split (open-loop proxy; closed-loop Lyapunov rollouts are future work), with the exact per-layer spectral deviation $\|W_{\text{comp}} - W_{\text{ref}}\|_2$ (power iteration) logged alongside. The predictor uses *only* weight-space features computable at compression time, is validated leave-one-out, and emits r^* with its measured counterpart; at the current point count the LOO R^2 is reported as indicative, not confirmatory. Artifacts: `results/lyapunov/{degradation_seed0.json, predictor.json}`.

C. Layer 3: Statistical tail bounds on formal robustness

Safety requirements are Signal-Temporal-Logic (STL) formulas with quantitative robustness ρ (Fainekos–Pappas semantics; $\rho > 0$ = satisfied with margin) [13]. The challenge metric “% episodes in a safe equilibrium” is the special case $\Pr[\rho \geq 0]$; we report the full ρ distribution and its tails.

TABLE II

COMPILER PASSES. PASS 4 IS AN IDENTITY, NOT AN APPROXIMATION.

Pass	Operation
1	Extract interaction table $\{(c_a, P_a)\}$
2	Group into mutually commuting families $\{F_1, \dots, F_m\}$ (greedy coloring)
3	Per family: single Clifford C_j with $C_j^\dagger P C_j \in \{I, Z\}^{\otimes k}$
4	Rewrite exactly $T = \sum_j C_j^\dagger D_j C_j$, D_j diagonal
5	Emit hardware table: qubits = $\lceil \log_2(\text{register}) \rceil$, depth = m , shots $\leq f(\ \sum_{a \in F_j} c_a \ _1)$
6	Emit pruning certificates: dropping F_j costs exactly $\ \sum_{a \in F_j} c_a \ _1$

[Measured in N7: predicted $\| \sum_{a \in F_j} c_a \|_1$ vs measured Δ]
Target: correlation ≥ 0.95 (E-comp-2); per-family leave-one-out

Fig. 3. Predicted vs measured pruning error — the ablation column no other submission can print.

The scored metric is estimated empirically by the classical arms; the Q-arm’s certified bound (Sec. V) is its conservative companion, stated with its direction.

IV. THE COMMUTING-PAULI INTERACTION COMPILER

Each cross-modal pairwise interaction term is a Pauli string $P_a = \sigma_{a_1} \otimes \dots \otimes \sigma_{a_k}$ with coefficient c_a (learned or inherited); the interaction tensor is $T = \sum_a c_a P_a$. Grouping into commuting families and per-family Clifford diagonalization are standard VQE measurement-reduction machinery [11], [12]; our contribution is their use as an exact, prunable compiler for cross-modal interaction tensors. The compile pass:

Exactness (E-comp-1): $\|T_{\text{exact}} - T_{\text{compiled}}\| \leq 10^{-6}$ relative (numerics/coefficient-estimation only). Runtime cost per family scales with the diagonal evaluation, not pairwise $O(N^2)$ enumeration — the attention bottleneck answered with an identity plus an $O(N \cdot k)$ -style diagonal kernel. Pruning is compositional with Eq. (3): the certified safe set survives a *documented* pruning budget.

Scope and complexity, stated honestly. The Pauli table is not extracted from an arbitrary dense matrix (that would require up to 4^k terms); it is extracted from the *compressed interaction accumulators* of Sec. VI, whose bounded rank bounds the term count, and families beyond a top- k by $\|c\|_1$ are pruned with a certificate-bound on the tail (R5). The grouping is greedy graph coloring — polynomial, not optimal — and the realized family count m is a *measured* output of N7, reported with its pruning certificate. We claim *no classical computational speedup* from the Clifford step: on classical hardware the rewrite is an exact algebraic identity whose value is exactness, certifiable pruning, and the hardware pathway; on quantum hardware, simultaneous measurement in a common eigenbasis is the standard VQE shot-reduction device that motivates commuting grouping, and k-NoCliD’s non-commuting relaxations [12] mark the adjacent frontier. The quantum content is the object language — Pauli observables, Clifford frame, measurement basis — not a speed claim.

Why this survives the QASA objection. QASA showed a capacity-matched classical low-rank bottleneck matches a PQC’s error metrics [9]. Our claim is not “the encoding is powerful” but “the compiler is exact, certifiable, and executable”: generic feature crosses have no commuting-group structure, and Performer/FAVOR+ [10] approximates the softmax kernel with distributional error and no notion of a family — it cannot prune with a printed certificate, and it emits no hardware table.

V. SAFETY EVALUATION ENGINE

A. Oracle

The Sec. 5.2 seeded-scenario requirement makes the simulator a reversible deterministic function of the seed:

$$O_{\text{true}} : \text{seed} \mapsto \text{scenario} \mapsto \text{rollout}(\pi) \mapsto \rho \mapsto [\rho < 0], \quad (6)$$

the object the classical arms evaluate. Amplitude estimation’s end-to-end value turns entirely on the oracle it can realize, so we state that explicitly. A coherent oracle that evaluates a *neural* rollout inside the circuit would carry policy and dynamics as reversible subcircuits — prohibitive depth on hardware and exponential state-vector cost on simulators. A coherent oracle that merely *loads* N pre-computed outcomes faces the $\Omega(N)$ state-preparation (QRAM) bottleneck. Our design avoids both horns:

- 1) *Structured domain, free preparation.* The seeded scenario set is a product domain (perturbation ranges per STL predicate, per the Sec. 5.2 sampling distribution), so the uniform superposition over $\log_2 |S|$ seed qubits is $H^{\otimes \log_2 |S|}$ — no amplitude-encoding of a lookup table. Reversible arithmetic for the predicates adds $O(\log |S|)$ ancillas, so the full AE circuit uses $O(\log |S|)$ qubits and simulated state-vector cost is polynomial in $|S|$ ($|S|^{O(1)}$) — tractable at our scenario counts under the challenge’s simulation mandate.
- 2) *Closed-form certificate-margin predicates.* The Q-arm oracle computes the *certified* verdict $\tilde{\rho}(s) = [\text{margin}_{\text{cert}}(s) < 0]$ (fusion joint 3): Layer-1 margins are arithmetic over core norms and box geometry; Layer-2a margins are the exported per-box SOS constants. Both are small reversible arithmetic/comparison circuits of printed size — never a neural rollout; the policy is deliberately outside the circuit.
- 3) *Conservative direction.* Sound certificates give $\Pr[\text{true failure}] \leq \Pr[\tilde{\rho}(s) < 0]$, so the Q-arm estimates an upper bound on the scored quantity — the correct ISO 26262 object, and the direction is stated on every reported number.

Oracle circuit size (qubits, depth, gates) and the query budget are printed per benchmark, so the claim is auditable: a *query-complexity advantage under an explicitly priced, realized oracle*. If the certificate stack degrades (R2/R3), the cheap-predicate premise degrades with it and the classical arms carry the section. Layer 3 is *offline evaluation-time*: the ≤ 100 ms real-time budget governs the runtime path of Sec. II (compiled diagonal kernels + monitor), which never executes an amplitude-estimation loop.

TABLE III
THREE-ARM SAFETY ESTIMATION (PRE-REGISTERED STRUCTURE;
COMPLEXITY AT EMPIRICAL $p = 10^{-5}$; THE Q-ARM’S ESTIMAND IS
 p_{cert}).

Arm	Method	Query cost
Q	IQAE, Bayesian posterior [14], [15]	$\sim 10^2$ – 10^3 oracle queries
C-strong	RESTART splitting + GEV tail fit [18]	10^3 – 10^4 rollouts (measured)
C-naive	Monte-Carlo counting	$\sim 10^6$ + (cost shown, not printed)

B. The three-arm race

Amplitude estimation estimates p to absolute error ε with $O(1/\varepsilon)$ *idealized-oracle* queries versus Monte-Carlo’s $O(1/\varepsilon^2)$ — quadratic query-complexity savings on the rarest scored quantity, standard in the QAE literature [14], [15], [17]. We claim no end-to-end speedup a priori: the race measures it. The Grover depth $1/\sqrt{p}$ is real simulator cost at our budgets — precisely why the race is pre-registered (R4) and the claim never assumes quantum hardware. The arms estimate complementary quantities: Q estimates the certified (conservative) violation probability p_{cert} on the cheap oracle above; C-strong estimates the empirical failure probability from true rollouts. R4 is defined on a common estimand: RESTART+GEV is also run on the *certified* predicate itself, so “Q loses” means IQAE’s credible interval on p_{cert} is wider than RESTART’s on the same quantity at matched total budget (printed oracle cost included); the empirical tail from true rollouts is reported alongside as the companion number. The crossover point (oracle complexity at which the classical arm wins) is a reported output of N6. If Q loses, we print that; Layers 1–2 stand independently. Adjacent QAE applications include structural reliability [15], flow-based failure sampling [16], and MPPI control [17]; Tabarraei and Das–Tanaka use lookup-table oracles over finite ensembles — precisely the case the oracle dilemma flags — and our in-domain delta is the realized cheap oracle over a structured domain fused with exact structural certificates.

Measured race (N6, this revision). On the seeded heavy-tailed margin oracle with analytic ground truth $p_{\text{true}} = 1.37 \times 10^{-4}$ (rare-event regime, per-seed deterministic rollouts), the three arms ran at matched query budgets: simulated Bayesian IQAE reaches a 95% interval width of 4.0×10^{-5} with ~ 630 oracle queries, while naive Monte-Carlo counting needs 10^6 queries for width 4.4×10^{-5} (a $\sim 1587\times$ budget ratio at matched width); the GEV extreme-tail arm covers the truth at 10^6 queries. All covering arms are reported with their coverage column; an arm that is narrow but misses p_{true} is scored *worse* than a wide honest one. Two scoping statements: (i) the RESTART *splitting* half of C-strong requires rollout trajectory access, which a seed $\rightarrow \rho$ oracle does not expose, so C-strong is implemented as the block-minima GEV estimator with block-bootstrap CI; (ii) the Grover amplification is *simulated* — the true amplitude lives only in the simulator and the estimator receives measurement counts exactly as it would on hardware. Scenario-optimization closure (Campi–Garatti,

$d = 2$, zero violations): $\varepsilon \leq 6.6 \times 10^{-4}$ at $N = 10^4$ ($\beta = 0.01$), $\varepsilon \leq 6.6 \times 10^{-6}$ at $N = 10^6$. Conformal coverage on fresh seeds: targets 0.99/0.95/0.90 measured as 0.9894/0.9475/0.8943. Artifacts: `results/safety/`.

C. Closure

Scenario optimization (Campi–Garatti): with N scenarios and solution complexity d , $\Pr[\text{violation} > \varepsilon] \leq \beta$ whenever $N \geq N(\varepsilon, \beta, d)$ — printed per claim as an (N, ε, β) triple [19]. **Conformal calibration**: all intervals receive split-conformal coverage correction on a held-out slice (never the training slice; leakage rule R12). **Falsification loop**: failures found by RESTART/IQAE feed back as training-slice augmentations, new long-tail slice entries, and monitor training positives; the report’s figure shows tail mass shrinking across rounds.

The runtime monitor enforces the certified-safe-set at inference time: at every step it checks that the model’s predicted action token lies within the certified ball of radius $L \cdot d$ around the reference action, where L is the per-layer Lipschitz product and d is the input-perturbation diameter of the scenario. If the predicted action falls outside the ball, the monitor returns a REFUSE_TO_SERVE sentinel and logs the event; otherwise it passes the action through. The interval logic is simple by design — its value is that it is (i) actually enforced, not merely proposed, and (ii) formally validated: `qicert.certify.guard.action_in_certified_set` is machine-checked against Z3 (B14g) for soundness (no out-of-ball action accepted), completeness (no in-ball action rejected), and range enforcement, and the guard’s manifest-based load-time path verifies the self-hash of every artifact’s manifest, the hash of every listed certificate file, and refuses service if any certified bound is NaN/Inf (B14h).

D. Certified-safe-set enforcement

At deployment, the monitor loads the signed manifest generated by `qicert.certify.manifest` at run completion, verifies the manifest’s self-hash and each artifact’s SHA-256, and then, per inference step, computes the predicted action token and checks

$$\text{accept} \iff 0 \leq a < N_{\text{actions}} \wedge |a - a_{\text{ref}}| \leq L \cdot d,$$

where a_{ref} is the certified reference action for that step (the frozen-eval action from the same frozen episode used to derive the certificate). The $L \cdot d$ radius is the product of the per-layer certificate bounds from the QTT-compressed model (Sec. III, Layer 1). If the bound is uncomputable (NaN/Inf) for any layer, the monitor refuses service — it never falls back to an uncomputable certificate.

The guard is deliberately minimal at the decision level; the trust comes from the verification of the artifacts that feed it and from the formal SMT check of the guard logic itself. A deployed monitor would additionally log each step’s a , a_{ref} , $L \cdot d$, and accept/reject decision to a tamper-evident log (the manifest’s self-signing pattern extends to the log).

End-to-end demonstration (this revision). `scripts/demo_guard.py` builds a certified run directory through the real certificate chain (TT-SVD $\rightarrow$ residual compensation $\rightarrow$ sound Lipschitz bound $\rightarrow$ signed manifest) and exercises both gates: a clean deployment is SERVED; an out-of-ball action is REFUSED (safe fallback engaged); a certificate tampered *after* manifest signing is REFUSED (hash mismatch); a NaN bound is REFUSED; a missing manifest is REFUSED. Transcript (1 serve / 4 refuse): `results/guard-demo/transcript.json`.

A two-layer guard on the action stream with theorem-budgeted guarantees.

Layer 1 — PDU hygiene (zero extra FLOPs). The compiled commuting-family table (Sec. IV, pass 3/6) already computes a syndrome per inference step. If the syndrome changes between consecutive steps, the packet is gray-listed: this catches bit-flip corruption of the Pauli register — sensor dropouts, camera blinding, CAN-bus corruption — the exact failure modes of the challenge’s synthetic perturbation suite.

Layer 2 — Classical-shadow statistics. With M random projection directions $v_1 \dots v_M$ (Huang–Kueng–Preskill shadows [20]), median-of-means concentration over the scalars $s_i(x) = \langle G_i, x \rangle$ gives a statistically correct false-alarm budget at memory $O(M \cdot s)$ and a few matvecs of latency — continuous OOD shift that a parity bit cannot see. A split-conformal gate sets the user-visible false-alarm rate (e.g. $\leq 1\%$). Gray-list fallback: the largest-radius action from the Layer-1 certified safe set — the monitor never aborts into the void.

VI. TRAINING ENGINE

QTT reshape [6]: $m = \prod m_i$, $n = \prod n_j$ (binary or small-factor modes, bit-ordering per layer type; N2’ resolves the ordering). **TT-cross** [5], [7]: cores $G_1 \dots G_d$ are built from $O(\sum m_i n_i r_i^2)$ maxvol queries of W ’s entries, never forming W — SVD-free and VRAM-safe for 7B-scale layers.

Expressiveness, measured not assumed. Long-range multimodal attention is known to be low-rank-friendly (softmax kernels compress well at moderate rank), but we do not assume it: N2 registers a per-layer rank-vs-accuracy curve at the target ratios with a pre-registered bound (R1: $> 5\%$ drop at $2\times$ demotes the compression claim), and the bond allocator then spends rank where the *certified long-tail margin* is cheapest in certificate terms.

ALS/DMRG-native fine-tuning: each core is optimized directly (regularized least squares with exact multilinear gradients; AMEn enrichment modulates bond dimension where the allocator’s safety signal flags it). Memory scales with core size, not layer size. Training passes: P1 baseline fine-tune; P2 compress $\rightarrow$ ALS fine-tune (the deliverable); P3 safety-critical last-pass on the RESTART/IQAE-falsified failure slice only (tail-shrinkage evidence, $\sim$ hundreds of hard examples).

Safety-budgeted bond allocator: per layer,

$$r^* = \arg \min_r \max_{\text{slices}} (\text{certified long-tail margin at } r), \quad (7)$$

TABLE IV
EXPERIMENT MATRIX (PRE-REGISTERED; GPU-H ARE QUOTA-HOURS).

ID	Experiment	Row	Pillar	GPU-h	cert-QTT	INT8	SV
N1	AD backbone fine-tune + INT8 reference	Accuracy @ 2× (drop vs dense)	A	8	$\leq 5\%^\dagger$	measured	meas
N2	{TT,QTT}×TT-cross sweep, 6 bond plans × 2 backbones	Accuracy @ 4× (drop vs dense)	A	24	target $\leq 5\%^\dagger$	measured	meas
N2'	Bit-ordering sensitivity (3 orders, 1 layer, 1 seed)	Certified-safe-set % @ 2×	A	2	$> 90\%^\dagger$	— (not computable)	—
N3	Layer-1 certificate table (Lipschitz products + tightness ratio κ)	Certified-safe-set % @ 4×	A	2	target $> 50\%^\dagger$	—	—
N4	Layer-2a SOS boxes (3 points × 2 tracks × slices)	Latency (ms, stated profile)	A	3	≤ 100	measured	meas
N5	Layer-2b Lyapunov curve + bond-spectrum predictor		A	6			
N6	Layer-3 safety suite (IQAE vs RESTART+GEV vs MC; oracle cost + crossover)		C	8			
N7	Compiler exactness + prune certificates + latency		B	6			
N8	(optional) PQC RL toy head + noise sweep		tertiary	4			
N10	Syndrome-shadow monitor demo	Metric	monitor	3		Seed 0 (mean ± std)	95%
N11	ALS/DMRG-native fine-tune, both backbones	engine	engine	8			
N12	Safety-critical last-pass on falsified slice	Fine-tuned action accuracy	engine	3			0.4453 ± 0.0001
N13	Cross-track mirror (N2/N3/N5 on second backbone)	INT8-of-fine-tuned (Korchao Int8WeightOnlyConfig, GPU)	engine	3			0.0000
N14	Clean-env end-to-end bench audit	INT8 vs FT drop (matched protocol, both legs, full split)	engine	3			

using (1) layer sensitivity to the certified-margin computation, (2) bond-spectrum entropy, (3) monitor hygiene bits. The long-tail slice is a design selector, not a report line.

VII. EXPERIMENTAL SETUP

Tracks and baselines (Sec. 5.4): AD — LLaVA-1.5-7B (Alpamayo-adjacent head) on nuScenes [21]/CARLA-style slices, vs. INT8 bitsandbytes; robotics — MiniVLA-class backbone (MIT code, HF checkpoints; Open X-Embodiment lineage [22]) on LIBERO-style slices, vs. INT8. Strong extra baselines: SVD-at-matched-ratio, tuned Performer at matched capacity, RESTART+GEV (not naive MC). **Hardware:** Kaggle 2×T4 (30 GPU-h/wk, 120 h ceiling), local RTX 5060 for small runs; all budgets are quota-hours, silicon-hours logged separately. **Discipline:** mean ± std over ≥ 3 seeds; every report table has a ledger row (research/experiments.md); every experiment has a pre-registered kill criterion.

Budget rollup: 87 GPU-h core, 33 h headroom vs. the 120 h ceiling; weeks 1–6 schedule in 08-experiments.md. Week 1: N1+N2'+N3 (the cheapest exposure of the two load-bearing facts: does the baseline behave, and does QTT compress with certificates attached?). The optional N8 PQC demo uses a quantum natural policy gradient head [23] for illustration only; it is cut first under quota pressure (R10).

VIII. RESULTS FRAMEWORK

Per challenge Sec. 5.5, all values are mean ± std over ≥ 3 runs. Tables below state the pre-registered targets; measured values land from the ledger and are inserted verbatim (no rounding-into-a-target). No number in this report is fabricated: each row is regenerable from the per-run run.json in results/ plus the ledger (results/ledger.csv).

Measured so far (this revision): the classical-energy N1v2 baseline (MiniVLA-1B, frozen 108-episode LIBERO spatial split, streamed eval, seed 0 of 3 seeds, 1200 LoRA-finetune steps at batch 2, local RTX 5060):

TABLE V
MAIN RESULTS TABLE (STRUCTURE; $\dagger$ = PRE-REGISTERED TARGET).

Row	Pillar	GPU-h	cert-QTT	INT8	SV
8	A	8	$\leq 5\%^\dagger$	measured	meas
24	A	24	target $\leq 5\%^\dagger$	measured	meas
2	A	2	$> 90\%^\dagger$	— (not computable)	—
2	A	2	target $> 50\%^\dagger$	—	—
3	A	3	≤ 100	measured	meas
6	A	6			
8	C	8			
6	B	6			
4	tertiary	4			
3	monitor	3		Seed 0 (mean ± std)	95%
8	engine	8			
3	engine	3			0.4453 ± 0.0001
3	engine	3			0.0000

A zero INT8-of-fine-tuned value on this fresh LoRA-fine-tuned model is expected and honest: naive weight-only INT8 without calibration collapses a fp16-fine-tuned qwen2.5-0.5B's action-token logits. The honest comparison for N2 is therefore *N2'-compressed (INT4 base + certified TT residual) vs. calibrated INT8 at matched compression ratio*, not naive weight-only INT8 — the N2' pipeline includes a Hessian-aware residual-fitting stage (GPTQ-intrinsic recipe + the QuaSAR-stable-pseudoinverse guard) that is the actual “INT8-competitive” arm. Seeds 1 and 2 of the N1v2 baseline were launched concurrently with this writing and will land in the ledger as they finish.

Measured compression status (full-split, validated harness, this revision). The eval harness was independently validated: loading the unmodified fine-tuned weights through the compressed-eval swap path reproduces the streamed reference to +0.0000 (0.4468, 6,496 batches, 114,497 action tokens). On that protocol, per-channel *calibrated* INT8 is near-lossless at $2.00\times$ (0.4466, $\Delta = -0.0001$) — which also proves the earlier 0.0000 naive-INT8 row was a calibration artifact, not a property of INT8 — while the certified TT+residual track at its best measured point holds 0.1529 at $2.86\times$ ($\Delta = -0.2939$), with all 168 per-layer certificates sound. The pre-registered R1 bar (some point at $\geq 2\times$ with $\Delta \geq -0.05$) is therefore **not met** by the uniform and Frobenius-residual arms; the activation-weighted fit (N2R-v2, GPTQ-style) and mixed allocation are the pre-registered next measurements, and the honest current claim is the certificate stack (the only compressed model with sound runtime certificates), not accuracy leadership. Every number in this paragraph regenerates from results/ run artifacts + the ledger.

IX. ABLATION AND QUANTUM JUSTIFICATION

Per challenge Sec. 5.5, the QI-isolating ablation is mandatory and is designed to be the strongest section of the report.

A1 (compression×certificate): at matched ratio, INT8 matches accuracy but cannot report the certificate column — the ablation shows the asymmetry directly (the number that cannot be computed is the win). The tightness ratio κ is

TABLE VI
COMPILER ACCEPTANCE (E-COMP) AND THREE-ARM SAFETY (PER SLICE).

E-comp	Target
1 exactness	$\ T_{ex} - T_{comp}\ \leq 10^{-6}$ rel.
2 prune-certificate corr.	predicted vs measured ≥ 0.95
3 latency	≤ 100 ms at full table
4 hardware self-consistency	compiled vs simulated within tolerance
Arm	Output
Q (IQAE)	credible interval on $p_{cert} = \Pr[\text{margin}_{cert} < 0]$ (conservative)
C-strong (RESTART+GEV)	profile-likelihood CI on p
C-naive (MC)	cost shown; Clopper–Pearson CI
Scenario-opt	(N, ϵ, β) triple
Conformal	coverage on held-out slice

reported per point: the certificate column is conservative-but-quantified, and Layer 2a closes the gap locally.

A2 (compiler vs encoding): per-family leave-one-out with predicted error $\|\sum_{a \in F_j} c_a\|_1$ vs measured Δ ; and compiler-vs-encoding: the exact identity vs a bare feature cross (shows the family structure is what buys exactness, not the encoding label). Tuned Performer at matched capacity is the classical ceiling; winning it is upside, losing it does not kill the claim (the primary axis is exactness + certificates + hardware table).

A3 (IQAE vs strong classical): the three-arm race at matched query budget; if Q loses, the report prints that (R4) and the claim rests on Layers 1–2 and the strong classical arm.

A4 (monitor): syndrome-shadow vs autoencoder baseline: correction rate at matched overhead; false-alarm rate vs the shadow theorem bound ($\leq 3\times$ or demoted, R7).

A5 (training engine): ALS-in-compressed vs dense-then-compress at matched ratio on the safety slice ($\leq 2\%$ worse or demoted, R6).

X. LIMITATIONS AND RISKS

Pre-registered kill/demote criteria (full register in 09-risks-kills.md): R1 compression accuracy $> 5\%$ drop at $2\times \rightarrow$ CompactifAI-style softer claim + certificate story; R2 Layer-1 bound too loose ($< 50\%$ certified safe set) $\rightarrow$ SOS becomes primary; R3 SOS SDP intractable $\rightarrow$ Lyapunov curve primary, SOS reported as “attempted, too slow”; R4 IQAE loses to RESTART $\rightarrow$ RESTART stands, IQAE = future work; R5 family count too large $\rightarrow$ top- k by $\|c\|_1$ with certificate-bound; R6 ALS-in-compressed $> 2\%$ worse $\rightarrow$ demote to post-processing; R7 shadow FPR $> 3\times$ theorem $\rightarrow$ PDU hygiene only; R8 allocator tradeoff $\rightarrow$ reported honestly; R9 cross-track non-comparable $\rightarrow$ pipeline claim only; R10 quota overrun $\rightarrow$ N8 cut first, then N13 reduced; R11 clean-env failure $\rightarrow$ report held; R12 conformal leakage $\rightarrow$ claim degraded until re-run.

Explicit framing disclaimers. We claim no classical speedup from the Clifford step (Sec. IV) and place amplitude estimation strictly offline with printed oracle costs (Sec. V); both are stated in-body so no stronger claim can be imputed.

No end-to-end advantage over classical estimators is claimed for IQAE a priori — the pre-registered race at matched total budget and its measured crossover decide (Sec. V). Runtime parity with classical baselines is an acceptance target, not an assumption: E-comp-3 (≤ 100 ms at the full compiled table) and the N14 clean-env bench are the measured evidence.

Three things that cannot kill the core claim: (1) the exact Lipschitz-product certificate — a theorem, not a benchmark; (2) compilation-as-identity + pruning certificate — a mechanism, not a number; (3) the three-layer safety methodology — a formal spec plus faced estimators plus conformal closure, not a score. The final table is built from whatever layers survive their criteria; the package always runs and reports something certified.

What is actually measured vs. pre-registered in this revision: the classical-energy N1v2 baseline (MiniVLA-1B, full frozen 108-episode LIBERO spatial split, stream-eval, 3 seeds $\times$ 1200 LoRA fine-tune steps at batch 2 on local RTX 5060; seed 0 is complete with fine-tuned accuracy 0.4468 (Wilson 95%, streamed over 6,496 batches) — see results/N1v2; the INT8-of-fine-tuned leg runs on torchao Int8WeightOnlyConfig (GPU) over the same 6,496 frozen batches, giving a matched-protocol $\Delta(\text{INT8} - \text{FT})$ per seed once seeds 1–2 complete (staged one at a time; EXECUTION-PLAN.md). The measured table now also includes the FT-anchor validation ($+0.0000$), the calibrated-INT8 comparator (0.4466 at $2.00\times$), and the certified TT+residual verdicts (0.1529 at $2.86\times$, 168/168 certificates sound) — see the Results Framework measured-status paragraph. No number in this revision is fabricated; each final table row is regenerable from results/N1v2/{run_id}/run.json + the ledger.

Certification framing: the STL specs map to ISO 26262 / IEC 62061 / ISO 10218 artifacts (hazard $\rightarrow$ safety goal $\rightarrow$ STL formula $\rightarrow$ robustness predicate $\rightarrow$ certified tail probability), and the certificate stack provides the evidence trail an assessor expects.

Scope generalization (explicit limitation). All compression measurements to date are on one backbone (MiniVLA-1B, qwen2.5-0.5B LLM) and one task suite (LIBERO spatial, frozen 108-episode split). The measured TT-truncation collapse at $\geq 2\times$ is partly a property of this backbone’s unusually flat weight spectra (information spread across singular directions; little discardable redundancy) and its fragile action head; literature reports structured and transformer backbones compressing far more gracefully. The certificate machinery, the guard, the estimator race, and the conformal closure are *model-agnostic by construction* (they operate on whatever weights the pipeline produces). Phase-2 work: repeat the full pipeline on a second backbone and a second task suite; extend Layer-2b to closed-loop Lyapunov rollouts; add an INT4 comparator for the $> 2\times$ regime; and a hardware-queue run. Layer-2a SOS boxes remain scoped behind the Julia pin (Q22) and are reported as attempted-not-run at submission if the pin does not land.

XI. RESOURCE DECLARATION

Hardware (quantum simulation tier): Kaggle 2×NVIDIA T4 16 GB (primary), local RTX 5060 laptop (secondary, development + classical baseline). **Hardware (classical-energy baseline tier):** local RTX 5060 (1×8 GB) — the N1/N2 classical-energy runs that validate the energy accounting and provide the QNN-vs-classifier fidelity context for Q3, all on this host only (no GPU cluster relied upon). **Total:** 87 GPU-h core (quota-hours) + headroom; silicon hours logged separately in the ledger. **Simulation stack (quantum):** CUDA-Q (pinned version), PennyLane, tntorch — versions in `environment.yml`; CUDA-Q available through Docker on this host. **Energy baseline stack (classical):** PyTorch 2.14 + torchao + peft + HF transformers 5.x + MiniVLA-1B backbone weights — versions in `docs/resource-declaration.md` (auto-generated from run provenance during the live bench). **Streamed eval workload note:** on the classical-energy baseline tier, the eval split is consumed lazily (two fresh stream iterators, one per eval leg) so the whole 108-episode held-out set (6,496 batches, 16 GB of batches in the earlier design) fits stably in 10 MB per batch without materialization — the earlier materialized-batch design OOM-killed processes silently; the streaming fix is in `bench/compress.py` since 2026-09-11. **Shot/gate budgets:** per quantum benchmark, stated in the compiler’s pass-5 table. **Energy:** measured by the `qicert` profiler (kWh + CO₂e at declared grid intensity). Reproducibility evidence: `qicert.bench.all` from a fresh virtualenv, verified by the certify suite (B14a–B14h) and `–skip-int8` handoff for N2.

XII. TRUST AUGMENTATION: MAKING EVERY NUMBER DEFENSIBLE

The single biggest failure mode in this kind of submission is an implicit or explicit claim that is not traceable to a defensible number — a number that a skeptical reader could not reproduce without asking a follow-up question. We mitigate this with a *trust-augmentation stack* (B14 family) that makes every reported number one of three things: measured with a confidence interval, checked against an independent verifier, or witnessed by a machine-checkable artifact. **No number in the final tables is claimed on the authority of the person who ran it.** The stack is pure-Python (no external formal-methods heavy-lifting required) and is developed and tested in this repository:

- **B14a — certifying-algorithm witnesses + property-based tests (Hypothesis).** Every certificate artifact ships as a `Witness` dataclass containing the serialized cores, the claimed Lipschitz float, and the bound-kind. An *independent* verifier re-derives the bound from the cores and asserts equality to tolerance. Property tests assert gauge-invariance of the TT contraction (bond rescaling with explicit inverse on a chain leaves the contracted operator invariant to machine precision), product-of-core-norms determinism, power-iteration monotonicity, *and* tamper detection: a witness whose shipped bound has been perturbed away from the true value must FAIL the verifier

— the witness is an integrity artifact, not a decoration. (See `tests/test_certify_properties.py`.)

- **B14b — statistical robustness (Clopper–Pearson).** The safety layer’s failure probability is a tail event ($p_{\text{cert}} = \Pr[\text{margin} < 0]$) and must be reported with a confidence interval, not a point estimate. We use two-sided Clopper–Pearson binomial intervals (the honest, conservative interval regulators expect) over the frozen 108-episode held-out split. Property tests verify interval containment, boundary behavior, and monotonicity in the violation count — so the reported interval is guaranteed to be at least as conservative as claimed. (See `tests/test_certify_properties.py`.)
- **B14c — interval/exact-arithmetic certificate cross-check.** The float pipeline’s certificate bound for a small layer is independently recomputed in 50-digit mpmath precision (power-iteration on $A^T A$ — dependency-free; this version fixes an earlier prototype that called the non-existent `mp.linalg.svdvals`) and, for small test layers, in exact `fractions.Fraction`. The float bound is accepted only if it matches the 50-digit recomputation to tolerance and, where conversion-safe, the exact fraction. This catches “your bound is a float artifact” attacks. (See `tests/test_certify_properties.py` + the `qicert.certify.cross_check` module.)
- **B14g — SMT falsification of the runtime guard (Z3).** Rather than proving the guard sound by hand, we ask Z3 to *disprove* its properties: the SMT encoding of `action_in_certified_set` is checked for soundness (no out-of-ball action accepted), completeness (no in-ball action rejected), and range enforcement. If Z3 found a counterexample, the guard would *not* be deployable; the unsat results in `tests/test_certify_guard.py` are the machine-validated assurance the guard is sound and complete on its stated domain. We also cross-validate the implementation against an independent Z3 model on 200 random (`action, reference, margin, n`) points.
- **B14h — signed manifests + runtime assertion guard.** At run completion, `qicert.certify.manifest.from_dir` writes a manifest that SHA-256-hashes every artifact (model weights, certificates, frozen eval split, provenance files) and self-signs by hashing the manifest itself. The runtime guard `qicert.certify.guard.check_certs_before_serve` re-verifies the manifest self-hash on load (tamper detection), verifies every listed artifact’s hash, and refuses service if any bound is NaN/Inf. Property tests verify tamper detection on real files (mutated artifact, forked manifest entry, missing artifact, NaN bound) — so the “receipts are enforced” narrative is real rather than aspirational.

The honest story. With the B14 stack, every number in the

final tables says: *this was measured with a confidence interval / checked against an independent verifier / witnessed by a machine-checkable artifact*, and the composition rule itself (the runtime guard) is machine-verified by a third-party SMT solver. That is stronger than most submissions’ entire verification story. The B14 stack is in development in this repository and is being refined as the quantum experiments land; it is architecture-complete (all of B14a–B14h implemented and tested) even where the quantum artifacts it validates are still pending.

XIII. CONCLUSION

We submit objects the baselines structurally cannot produce: exact certificates that are functions of the compressed cores, an exact compilation identity with per-family provable pruning, quadratic query-complexity savings on the rarest scored quantity under a realized, priced oracle, and a package that reproduces every table from a clean environment. All claims are either theorems, structural properties, or pre-registered measurements with kill criteria — the honesty contract of Sec. 5.5 operationalized. The full argument, theory, and risk register are maintained in the submission folder of the qicert repository.

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